

Running Models Using OLLAMA:

tinyllama-1.1b-chat-v1.0.Q5_K_M.gguf

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botic@embotics:~ $ python3 untitled.py
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Loading TeacherBot via Ollama (teacherbot-tinyllama)...
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Model loaded successfully.
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```
Load Time : 50.03 sec
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Script RAM Usage : 32.9 MB
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Ask TeacherBot (type 'exit' to quit):
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Ask TeacherBot (type 'exit' to quit): what is physics
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TeacherBot
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Physics is a branch of science that deals with the study of the natural phenomena and the properties of matter and energy. It involves the understanding of the fundamental principles of motion, force, pressure, heat, and sound, among others. Physics is essential in many fields such as engineering, medicine, physics education, and technology.
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Physics is divided into several subfields, including:
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1. Mechanics: This subfield deals with the study of motion and forces acting on objects. It includes topics like Newton's laws of motion, force, and energy, as well as topics like fluid mechanics, dynamics, and kinematics.
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2. Electromagnetism: This subfield deals with the study of electrical and magnetic fields, currents, and electric and magnetic forces. It includes topics like Maxwell's equations, electrodynamic

Performance Metrics

Prompt Tokens : 115

Generated Tokens : 180

Total Tokens : 295

Response Length : 816 characters

Generation Time : 27.99 sec

Wall Clock Time : 32.70 sec

Tokens / Second : 6.43

Script RAM Usage : 32.9 MB

Ask TeacherBot (type 'exit' to quit): how a person can reach mars

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TeacherBot

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To reach Mars, there are several methods that individuals can use:

1. Spacecraft: The most common method for reaching Mars is through spacecraft. NASA's Mars missions, such as the Mars Reconnaissance Orbiter and the Mars Odyssey, have been successful in sending probes to the Red Planet. These missions have provided valuable data about the planet's geology, atmosphere, and surface features.

2. Rovers: NASA also sends rovers to explore Mars, such as the Curiosity Rover, which landed on Mars in 2012. The rover has been able to collect samples of Martian soil and rock, which have provided valuable information about the planet's geology and environment.

3. Landing on Mars: Finally, individuals can also attempt a landing on Mars by using a

Performance Metrics

Prompt Tokens : 118

Generated Tokens : 180

Total Tokens : 298

Response Length : 746 characters

Generation Time : 27.63 sec

Wall Clock Time : 28.27 sec

Tokens / Second : 6.52

Script RAM Usage : 33.0 MB

Ask TeacherBot (type 'exit' to quit):

qwen2.5-1.5b-instruct-q6_k.gguf

botic@embotics:~ \$ python3 untitled.py

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Loading TeacherBot via Ollama (teacherbotQwen0.5)...

=====

Model loaded successfully.

Load Time : 8.43 sec

Script RAM Usage : 32.9 MB
=====

Ask TeacherBot (type 'exit' to quit): what is physics

=====

TeacherBot

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Physics is a fundamental science that deals with the study of matter and energy in the physical world. It encompasses a wide range of topics including:

1. **Mechanics**: Deals with the motion, forces, and interactions between objects.
2. **Thermodynamics**: Studies heat, work, and energy transfer processes within systems at constant temperature.
3. **Electromagnetism**: Explains how electric charges interact and generate magnetic fields.
4. **Fluid Dynamics**: Focuses on the flow of fluids (liquids or gases) in a medium.
5. **Optics**: Deals with the study of light, including its behavior, reflection, refraction, and polarization.
6. **Calculus**: Essential for understanding motion, change, and rates of change.
7. **Quantum Mechanics**: Explains the behavior of matter at extremely small scales, such as atoms and subatomic particles.

Performance Metrics

Prompt Tokens : 88

Generated Tokens : 180

Total Tokens : 268

Response Length : 853 characters

Generation Time : 17.57 sec

Wall Clock Time : 21.15 sec

Tokens / Second : 10.25

Script RAM Usage : 32.9 MB

Ask TeacherBot (type 'exit' to quit): how a person can reach mars

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TeacherBot

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To reach Mars, you would need to use advanced technology and spacecraft designed for such missions. Here's a step-by-step guide:

1. **Planning**: First, you would plan your mission using various tools like NASA's Mars Exploration Program or other space agencies' plans.

2. **Launch**: Once the planning is complete, you would launch your spacecraft into space using rockets. This involves sending a rocket to Mars and then having it reach Earth orbit before reaching Mars.

3. **Orbital Transfer**: After reaching Earth orbit, your spacecraft would need to perform an orbital transfer maneuver to enter Mars' orbit. This involves moving the spacecraft from one point in its orbit to another.

4. **Mars Orbit**: Once in Mars' orbit, you would use a series of maneuvers to get closer and closer to Mars. This could involve using gravity assists from other planets or asteroids.

5. **Capture**

Performance Metrics

Prompt Tokens : 91
Generated Tokens : 180
Total Tokens : 271
Response Length : 892 characters
Generation Time : 16.91 sec
Wall Clock Time : 17.47 sec
Tokens / Second : 10.65
Script RAM Usage : 33.0 MB

Ask TeacherBot (type 'exit' to quit):

qwen2.5-0.5b-instruct-q6_k.gguf

botic@embotics:~ \$ python3 untitled.py

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Loading TeacherBot via Ollama (teacherbotQwen0.5)...

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Generated Tokens : 180

Total Tokens : 271

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Generation Time : 16.91 sec

Wall Clock Time : 17.47 sec

Tokens / Second : 10.65

Script RAM Usage : 33.0 MB

Ask TeacherBot (type 'exit' to quit):

Running Models Using LLAMA-CPP

[tinyllama-1.1b-chat-v1.0.Q5_K_M.gguf](#)

Loading TeacherBot...

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Model loaded successfully.

Load Time : 1.22 sec

RAM Usage : 1470.7 MB

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Ask TeacherBot (type 'exit' to quit): what is physics

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TeacherBot

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Physics is the branch of science that deals with the study of motion, forces, and the structure and behavior of matter. It involves the understanding of the principles of motion, energy, and matter, as well as their interactions. Physics covers a wide range of topics, including:

1. Motion: The study of how objects move, such as in the movement of planets, stars, and galaxies.
2. Forces: The study of the forces that cause motion, such as gravity, electromagnetism, and nuclear forces.
3. Structure and behavior: The study of the properties of matter at different scales, including atoms, molecules, and solids.
4. Energy: The study of how energy is transferred between objects or systems, such as heat, electricity, and light.

Physics is an important branch of science

Performance Metrics

Prompt Tokens : 114

Generated Tokens : 180

Total Tokens : 294

Response Length : 776 characters

Generation Time : 35.53 sec

Tokens / Second : 5.07

RAM Usage : 1487.1 MB

Ask TeacherBot (type 'exit' to quit): how a person can reach to mars

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TeacherBot

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To reach Mars, a person would need to follow several steps:

1. Launch a rocket from Earth's surface.
2. Once the rocket is in space, it will use its engines to accelerate towards Mars.
3. The rocket will enter Mars' atmosphere and begin to slow down.
4. After entering Mars' atmosphere, the rocket will use its parachutes to descend towards the planet's surface.
5. Once the rocket has reached a safe distance from Mars, it will deploy its landing legs and use them to touch down on the Martian soil.

To reach Mars, a person would need to follow these steps with precision and accuracy. The rocket would need to be designed and built carefully to ensure that it is capable of reaching Mars safely. The rocket's engines would need to be powerful enough

Performance Metrics

Prompt Tokens : 118

Generated Tokens : 180

Total Tokens : 298

Response Length : 753 characters

Generation Time : 33.25 sec

Tokens / Second : 5.41

RAM Usage : 1487.4 MB

Qwen2.5-0.5B-Instruct 32,768 Q6_K ~506 MB

Loading TeacherBot...

=====
Model loaded successfully.

Load Time : 1.21 sec

RAM Usage : 1057.6 MB

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Ask TeacherBot (type 'exit' to quit): what is physics

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TeacherBot

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Physics is the branch of science that deals with the study of matter, energy, and their interactions. It covers a wide range of topics including:

1. **Mechanics**: This involves studying motion, force, acceleration, and other physical properties of objects.

2. **Kinematics**: This focuses on describing the motion of an object without considering its mass or direction. Examples include motion in one dimension (linear motion) and two dimensions (cartesian coordinates).

3. **Dynamics**: This deals with the study of forces acting on a body, including Newton's laws of motion.

4. **Electromagnetism**: This includes the study of electricity and magnetism, focusing on electric fields, currents, and magnetic fields.

5. **Thermodynamics**: This studies heat transfer, work, energy transformations (like chemical reactions), and entropy.

6. **Fluid Dynamics**: This is about fluids

Performance Metrics

Prompt Tokens : 88

Generated Tokens : 180

Total Tokens : 268

Response Length : 886 characters

Generation Time : 36.04 sec

Tokens / Second : 4.99

RAM Usage : 1068.5 MB

Ask TeacherBot (type 'exit' to quit) how a person can reach to mars

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TeacherBot

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To reach Mars, a person would need to follow several key steps:

1. **Spacecraft Development**: First, the spacecraft needs to be developed and built. This involves designing and building the vehicle that will carry humans to Mars.
2. **Launch Preparation**: Once the spacecraft is ready, it must be prepared for launch. This includes planning the launch schedule, ensuring all safety protocols are in place, and conducting any necessary tests or inspections.
3. **Orbiting the Earth**: The spacecraft needs to orbit around Earth to reach a stable orbit that allows for the launch of the Mars mission. This involves using gravitational forces from Earth to help guide the spacecraft towards Mars.
4. **Mars Landing**: Once in orbit, the spacecraft will attempt to land on Mars. This can be done through various methods such as parachutes or rockets, depending on the design of the spacecraft and the specific landing conditions

Performance Metrics

Prompt Tokens : 92
Generated Tokens : 180
Total Tokens : 272
Response Length : 930 characters
Generation Time : 36.27 sec
Tokens / Second : 4.96
RAM Usage : 1073.8 MB

qwen2.5 1.5B instruct Q2-k

botic@embotics:~ \$ python3 ~/llama.py

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Loading TeacherBot...

=====
Model loaded successfully.

Load Time : 0.89 sec

RAM Usage : 839.5 MB
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Ask TeacherBot (type 'exit' to quit): what is physics

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TeacherBot

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Physics is the study of matter and its behavior under different conditions. It deals with objects and their arrangement in space, how they interact with each other, and what happens to them as a result. Physics helps us understand the world around us by using mathematical equations and scientific principles. It's like studying the rules behind things that move and stick together.

Performance Metrics

Prompt Tokens : 88

Generated Tokens : 69

Total Tokens : 157

Response Length : 382 characters

Generation Time : 18.84 sec

Tokens / Second : 3.66

RAM Usage : 854.9 MB

Ask TeacherBot (type 'exit' to quit): how a person can reach mars

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TeacherBot

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A person can reach Mars by traveling to it. This journey involves a spacecraft, which is sent out into space and travels at the speed of light (1075 kilometers per second). If you multiply this speed by 265, you get how many seconds in one year: 315,846,900. Multiply that by 60 to find how many minutes it takes: 1,120,588,200. Divide the number of minutes with 24 and add 1 for a minute: 447,315,000. Subtract this from the original time: 315,439,000. Take that plus 60% off to find how many days it takes: 5,875,820. Divide that by

Performance Metrics

Prompt Tokens : 91

Generated Tokens : 180

Total Tokens : 271

Response Length : 534 characters

Generation Time : 39.07 sec

Tokens / Second : 4.61

RAM Usage : 856.7 MB

Ask TeacherBot (type 'exit' to quit):